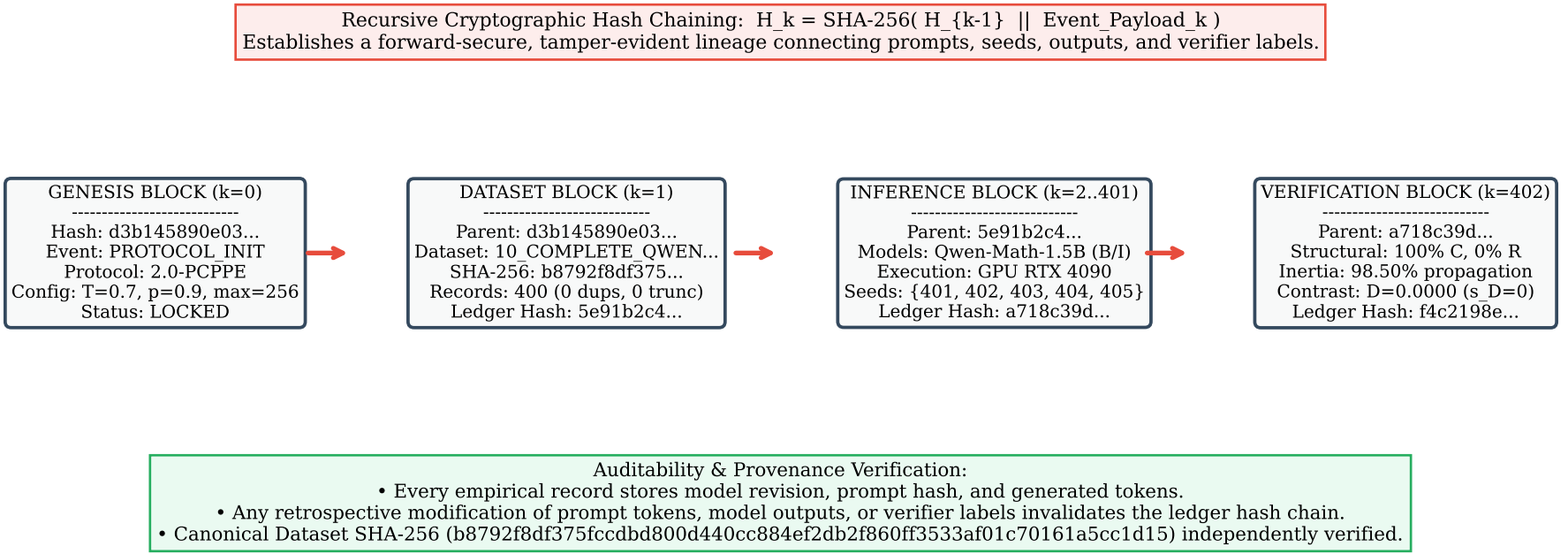
Recursive Cryptographic Hash Chaining: H_k = SHA-256(H_{k-1} || Event_Payload_k)
Establishes a forward-secure, tamper-evident lineage connecting prompts, seeds, outputs, and verifier labels.
GENESIS BLOCK (k=0)
Hash: d3b145890e03...
Event: PROTOCOL_INIT
Protocol: 2.0-PCPPE
Config: T=0.7, p=0.9, max=256
Status: LOCKED
DATASET BLOCK (k=1)
Parent: d3b145890e03...
Dataset: 10_COMPLETE_QWEN...
SHA-256: b8792f8df375...
Records: 400 (0 dups, 0 trunc)
Ledger Hash: 5e91b2c4...
INFERENCE BLOCK (k=2..401)
Parent: 5e91b2c4...
Models: Qwen-Math-1.5B (B/I)
Execution: GPU RTX 4090
Seeds: {401, 402, 403, 404, 405}
Ledger Hash: a718c39d...
VERIFICATION BLOCK (k=402)
Parent: a718c39d...
Structural: 100% C, 0% R
Inertia: 98.50% propagation
Contrast: D=0.0000 (s_D=0)
Ledger Hash: f4c2198e...
Auditability & Provenance Verification:
• Every empirical record stores model revision, prompt hash, and generated tokens.
• Any retrospective modification of prompt tokens, model outputs, or verifier labels invalidates the ledger hash chain.
• Canonical Dataset SHA-256 (b8792f8df375fccdbd800d440cc884ef2db2f860ff3533af01c70161a5cc1d15) independently verified.